Introduction

A significant portion of the world's population consists of children and neonates. Millions of children suffer from diseases that either do not have cures or have medicinal cures that are tailored to adults (Petkova.et al., 2023). Inaccessibility to pediatric drugs is an issue that prevents young patients from obtaining medication that is tailored to their bodies. There are several parties involved in this issue aside from the pediatric patient themselves, such as parents, healthcare providers, researchers, pharmaceutical industries, and the government. The result of this issue is the use of off-label drugs, manipulation of drugs that were researched and manufactured for adults, and the overall lack of funding dedicated to making age-appropriate treatments for children.

To provide care for ailing children, certain alternative solutions have been derived based on the situation (Bourgeois et al., 2012). As mentioned above, one of these alternatives is the prescription of off-label drugs. The term "off-label" drugs refer to any clinically prescribed medication that has not yet been approved by the Food and Drug Administration (FDA) (Allen et al., 2018). Due to the lack of medications specifically tailored for young children, physicians are often forced to prescribe such medication when no other alternative is available. Although this is not illegal, as the drug is actively being researched, the lack of knowledge of how the drug would work in a child's body poses a significant risk for adverse effects manifesting in the patient (Bourgeois et al., 2012).

Due to the lack of age-appropriate medication for children, another alternative for providing care is the manipulation of drugs that were developed for adults. This is usually done by breaking a tablet in half or dissolving the contents of a capsule in water (Petkova et al., 2023). The issue with this alternative is that along with the risk of dosage error, drugs that were researched and formulated for adults may not work in the same way for children due to physiological differences (Kern, 2009).

Such alternatives usually work, however, the threats they pose show the necessity of an increase in clinical trials that are tailored to pediatric drugs. Pediatric clinical trials get overlooked for reasons involving economics and ethics. There are a lot of barriers that make recruiting underage personnel for clinical trials difficult, which makes researchers focus on trials where they can easily recruit adults (Lagler et al., 2021). Furthermore, the market for adult prescription drugs is more profitable than that of pediatric drugs. Therefore, pharmaceutical companies tend to provide funding for research that would yield profitable results (Singh et al., 2025).

This literature review will go into further detail regarding the barriers that pediatric patients face in accessing well-researched, age-appropriate drugs that are on the market.

Lack of Clinical Trials in Pediatric Drug Development

Children have been denied new medication due to lack of proper clinical trials, and the literature confirms that this continues now, despite decades of regulatory attention. Several converging factors economic disincentives, structural and ethical barriers, and unresolved methodological gaps help explain why pediatric clinical trials remain much lower in comparison to adult trials. Pediatric trials have historically received less priority because of financial and marketing considerations, while small eligible populations and the complexity of obtaining parental consent make these trials more difficult to conduct than adult trials (Caldwell et al., 2004).

Singh et al. (2025) attributes much of this gap to economics: pediatric drug development remains underfunded relative to adult drug development, and several pharmaceutical companies have recently closed dedicated pediatric drug development units to focus on more profitable adult markets. This is not a new problem Naka et al. (2017) found that although both pediatric and adult trial volumes have grown over recent decades, pediatric growth has lagged substantially behind adult growth, with physiological differences (illustrated through dermatologic conditions) frequently overlooked when adult trial data is extrapolated to children by default.

The evidence shows that this problem is continues. In a cross-sectional analysis of 1,088 pediatric randomized controlled trials (RCTs) registered between 2011 and 2013, Nebeska et al. (2025) found that 16.5% were never completed, mainly because of bad patient recruitment, and even trials that were completed were often not published with a median 21-month delay between trial completion and published results. Furthermore, among 559 pediatric randomized controlled trials registered from 2008 to 2010, 19% were discontinued early, most commonly because of difficulty recruiting participants, and 30% of the 455 completed trials remained unpublished (Pica & Bourgeois, 2016).

Domingues et al. (2023) also explain that pediatric drugs are harder to develop because the dosage form, taste, and method of administration must be suitable for children. They note that even with US and EU laws supporting pediatric drug development, newer treatments such as nanomedicine are still slow to become available for children.

Lagler et al. (2021) noted the structural barriers: pediatric populations are smaller and more heterogeneous than adult populations, trials require both parental consent and child agreement, and a single trial protocol must often account for physiological differences across infants, and adolescents, making international pediatric trial networks important because they help bring together enough patients for research.

These structural challenges are not new. Kern (2009) observed that even after legislation formally obligated developers to study medications in children, clinicians continued to default to empirical or experiential dosing rather than pediatric-trial-derived dosing a pattern later sources confirm has not meaningfully changed.

Carpenter et al. (2017) extend this into research ethics and methodology, summarizing a multidisciplinary conference that identified five priority areas for pediatric medication safety research, and noting that electronic medical record (EMR) data often proposed as an alternative to trial evidence is frequently inconsistent and incomplete, limiting its use as a genuine substitute.

FDA reviewers used complete or partial extrapolation of efficacy from adult or other data for 137 of 166 products, showing that extrapolation is a common component of pediatric development rather than evidence that adult and pediatric responses are always identical (Dunne et al., 2011). Among 125 drugs receiving pediatric labeling through FDA submissions from 2015 to 2020, complete or partial efficacy extrapolation was used for 74%, indicating that carefully justified extrapolation can help expand pediatric labeling while targeted pediatric dosing and safety evidence is collected (Samuels et al., 2023).

Even when trials are initiated, enrollment remains an obstacle. Bencheva et al. (2024) reviewed systematic reviews of pediatric trial enrollment barriers and found that the barriers that contribute to enrollment were well documented but there is limited evidence that the suggested recruitment strategies work. O'Hara et al. (2020) examined pharmacokinetic (PK) studies in infants specifically, looking into barriers including ethics approval, PK study design constraints, and the physical difficulty of drawing sufficient blood volume from small patients meaning infant dosing today is still largely extrapolated from adult and general pediatric PK data rather than derived from infant-specific studies.

Taken together, these sources show that the shortage of pediatric clinical trials stems not from a single cause but from a stack of reinforcing economic, structural, ethical, and technical barriers. A recurring thread of clinicians defaulting to adult-extrapolated dosing when pediatric trial data does not exist (Kern, 2009; Lagler et al., 2021; O'Hara et al., 2020) is the direct link into why off label prescribing is so common in pediatric care.

Inaccessibility to Age-Appropriate Pediatric Medications

Age-appropriate drug access remains a major challenge in pediatric care, especially for infants and neonates. Many existing formulations are unsuitable for children, frequently requiring clinicians to prescribe adult medicines off-label or unlicensed and manipulate their dosage forms before administration (Ivanovska et al., 2014). Polysorbates, for example, are common excipients in biotherapeutics and help protect protein drugs from interacting with surrounding layers. Although polysorbates are considered safe within recommended limits, they degrade through hydrolysis. The scoping review by Kriegel, C. (2020) highlights the gap in safety data between adults and children. Across 89 neonatal units in 21 European countries, 638 of 2,095 prescriptions contained at least one potentially harmful excipient, resulting in exposure among 456 of the 726 neonates studied (Nellis et al., 2015). Until 2006, half of the drugs given to children had never been properly examined in pediatric populations (Kriegel, C., et al., 2020).

There are also major knowledge gaps in pediatric pharmacokinetics and pharmacodynamics, especially because children are continuously growing and developing. This makes it difficult to predict how drugs move through and act within their bodies. The lack of reliable biomarkers further limits our ability to assess drug effects accurately. The review by Watanabe, H. et al. (2024) suggests that integrating pharmacogenetic information, such as how a child’s genes influence drug response, is essential for improving pediatric drug development and therapy.

Across 128 meta-analyses comparing adult and pediatric trials, researchers found no major overall difference in treatment effects. However, the data still showed meaningful discrepancies in how effective treatments were for children compared to adults (Contopoulos-Ioannidis et al., 2010). Another meta-analysis by Lathyris D. (2014) examined trials across seven categories of harm. This study showed that adult safety data cannot simply be applied to children, as significant differences were found in headaches and several specific harms (Lathyris D., et al., 2014).

Several regulatory programs in the United States were created to improve evidence-based medicine for children, yet many approved oral medications remain unsuitable for younger age groups. Out of 214 oral formulations evaluated, fewer than one third were appropriate for neonates. Most pediatric products are tablets or capsules, which are difficult for young children to swallow. Age-appropriate solid formulations have been shown to be more feasible, as 4-mm minitablets were more acceptable to healthy infants and preschool children than the syrups, suspensions, and powders evaluated in the same study (van Riet-Nales et al., 2013). Some liquid formulations also contain inactive ingredients added to improve drug stability or taste, but these excipients can pose health concerns for children (Weiler HR, et al., 2026).

Neonates undergo constant maturation and differentiation, which affects their pharmacokinetics and drug responses. This increases the need for formulations specifically designed for newborns. Bavdekar, S. (2013) discusses the ethical challenges that arise in neonatal research. Parents often need to make decisions during stressful moments immediately after birth, making consent difficult. Antenatal or opt-out consent approaches are sometimes recommended to reduce parental pressure, but these challenges contribute to the limited availability of drugs suitable for neonates and infants (Bavdekar, S., 2013).

Overall, the evidence shows that age-appropriate drug access to pediatrics is still far from adequate. Many medicines used in infants and neonates were originally designed for adults, which creates safety concerns with excipients, dosing, and formulation. Large gaps in pharmacokinetic, pharmacodynamic, and pharmacogenetic knowledge make it difficult to predict how children will respond to treatment, and existing trials often reveal meaningful differences in both efficacy and harms between adults and children. Even with regulatory efforts, a significant number of pediatric formulations remain unsuitable for younger age groups, especially neonates. Ethical challenges in neonatal research further limit progress, as consent, enrollment, and trial design are complicated by the vulnerability of newborns and the stress placed on parents. Together, these issues highlight the urgent need for better pediatric-specific drug development, safer formulations, and research practices that truly reflect the unique needs of children.

The Role of Off-Label Prescribing in Limiting Pediatric-Specific Drug Access

Many drugs used today for pediatric patients are prescribed outside their approved age and dosage or are not licensed for use in children by the approved regulatory bodies and are termed “off-label” medication (Petkova et al., 2023). Although off-label prescribing also occurs in adults, it is particularly prevalent among infants and children. A systematic review of 31 studies reported pediatric off-label prescribing rates ranging from 3.2% to 95% (Allen et al., 2018). Across studies published between 1996 and 2016, pediatric off-label prescribing rates ranged from 1.2% to 99.7%, with age and dose being the most common reasons for off-label classification and persistently high use occurring in neonatal intensive care units (Balan et al., 2018). Furthermore, a prospective study of children aged 4 days to 16 years in hospitals in the United Kingdom, Sweden, Germany, Italy, and the Netherlands found that 46% of 2,262 prescriptions were off-label or unlicensed and that 67% of the 624 children received at least one such prescription (Conroy et al., 2000). This prevalence of off-label prescriptions is especially found to be high within newborns receiving intensive care, with 96.4% of the 220 newborns studied in a neonatal intensive care unit received at least one off-label medication, while 49.3% of all medication items prescribed were off-label (Costa et al., 2018). International evidence reviewed across pediatric settings showed that off-label and unlicensed prescribing was widespread, particularly among newborns and hospitalized children, while pediatric safety information remained limited (Cuzzolin et al., 2006). These studies suggest that off-label prescribing is not a rare plan of treatment used under exceptional circumstances but rather a routine response to the limited availability of approved pediatric drugs.

The high prevalence of off-label prescriptions is also connected to the lack of accessible and age-appropriate medications for children. Pediatric patients cannot simply be treated as small adults because of their rapidly developing biology. Developmental changes in physiology alter drug absorption, distribution, metabolism, and excretion throughout infancy and childhood, meaning that pediatric dosing cannot be determined reliably by simply reducing an adult dose according to body size (Kearns et al., 2003). However, conducting high-quality pediatric clinical trials are complicated by ethical concerns and hindered by shortage of participants due to parental consent and high physiological differences between age groups (Meng et al., 2022). In some cases, when a pediatric drug is approved, it may not be commercially available in the proper formulation. For instance, Litalien et al. (2020) in Canadian pediatric hospitals, identified 56 medications that had to be compounded into liquid formulations for children where 27 of these medications already had suitable age-appropriate formulations available commercially outside of Canada. Among 270 relevant new drugs approved by Health Canada between 2007 and 2016, only 75 monographs contained a pediatric indication and only 7 contained a neonatal indication; moreover, just 9 of the 15 oral drugs indicated for children aged six years or younger were available in child-friendly dosage forms (Raja et al., 2020). Similarly, in countries such as the Netherlands it has been found that approximately only half of available medicines were suitable for at least one pediatric age group (van Riet-Nales et al., 2011). Therefore, access to pediatric drugs involves many barriers beyond regulatory approval which feed into lower availability and the increase of off-label reliance.

While some medications are termed “off-label”, this status does not automatically mean that a medication is ineffective or dangerous. Nevertheless, a prospective cohort study found that off-label or unlicensed medicines were more likely than authorized medicines to be implicated in adverse drug reactions, although this association was largely driven by medications used in oncology (RR = 1.67, 95% CI 1.38-2.02; Bellis et al., 2014). The American Academy of Pediatrics recognizes that off-label prescribing can be appropriate when it is based on scientific evidence (Neville et al., 2014). Furthermore, it is argued by researchers that off-label treatment can be appropriate when the expected benefits outweigh the risks to the patient (van der Zanden et al., 2021). Some off-label drugs have become accepted standard prescriptions when no age-appropriate alternatives exist. In these situations, physicians may rely on published research and clinical experience to select an effective treatment for their patients (Petkova et al., 2023).

Although the practice of using off-label medications provides children with the necessary treatment, the effectiveness and widespread acceptance of off-label medications also contribute to the current lack of pediatric specific medications. When physicians can address a patient's disease immediately with the use of a familiar off-label drug, it becomes harder to notice the need for specific pediatric approved medications. Due to this, an unintentional decrease in demand for pediatric drugs reduces incentive for pharmaceutical companies to conduct pediatric clinical trials (Milne & Bruss, 2008). While the overall availability of effective off-label alternatives is not the sole cause of limited pediatric drug developments and access but rather a reinforcing factor for the normalization of using off-label medication which allows the shortage of pediatric specific products to remain.

Nevertheless, clinical experience with certain off-label drugs cannot replace clinical studies and approval. It has been found that only 14% of 2718 studied off-label pediatric drugs were supported by high-quality evidence from randomized controlled trials (van der Zanden et al., 2022). Similarly, literature involving 212 off-label drugs showed that only 40 had FDA approval in the US (Meng et al., 2022). Therefore, although effective off-label prescribing can temporarily address patient needs, it ultimately reflects the lack of pediatric-specific drug development. Reliance on off-label drugs results in uncertainty in dosing and safety for both short and long-term patient outcomes and contributes to the continued lack of clinically tested and verified medication for pediatric patients.

Conclusion

To conclude, this literature review has identified significant issues with the process of accessing pediatric drugs. Though these issues span several disciplines, including the usage of off-label drugs due to the lack of age-appropriate alternatives and clinical trials, solutions can be developed to ensure that pediatric patients receive the treatments that are developed specifically for their bodies. Next steps include identifying solutions that can be implemented while the knowledge gaps of pediatric medicine are addressed. One such solution proposed by Dean et al. (2021) is to add the role of a Drug Access Navigator (DAN) to the patient's care team. A DAN would be able to increase the efficiency of obtaining drugs through government programs, while reducing the costs for the patient's family (Dean et al., 2021). Overall, as has been proven in several studies, if pediatric patient-specific drugs are developed and made more accessible to the public, several children can have more tailored treatments that can cure several types of rare diseases.

References

Allen, H. C., Garbe, M. C., Lees, J., Aziz, N., Chaaban, H., Miller, J. L., Johnson, P., & DeLeon, S. (2018). Off-label medication use in children, more common than we think: A systematic review of the literature. *The Journal of the Oklahoma State Medical Association, 111*(8), 776–783. https://pmc.ncbi.nlm.nih.gov/articles/PMC6677268/

American Academy of Pediatrics Committee on Drugs. (2014). Off-label use of drugs in children. *Pediatrics, 133*(3), 563–567. https://doi.org/10.1542/peds.2013-4060

Balan, S., Hassali, M. A. A., & Mak, V. S. L. (2018). Two decades of off-label prescribing in children: A literature review. *World Journal of Pediatrics, 14*(6), 528–540. https://doi.org/10.1007/s12519-018-0186-y

Bavdekar, S. B. (2013). Pediatric clinical trials. *Perspectives in Clinical Research, 4*(1), 89–99. https://doi.org/10.4103/2229-3485.106403

Bellis, J. R., Kirkham, J. J., Nunn, A. J., & Pirmohamed, M. (2014). Adverse drug reactions and off-label and unlicensed medicines in children: A prospective cohort study of unplanned admissions to a paediatric hospital. *British Journal of Clinical Pharmacology, 77*(3), 545–553. https://doi.org/10.1111/bcp.12222

Bencheva, V., Mann, N.-K., Rombey, T., Pieper, D., & Schmiedl, S. (2024). Barriers and facilitators to enrollment in pediatric clinical trials: An overview of systematic reviews. *Systematic Reviews, 13*, Article 283. https://doi.org/10.1186/s13643-024-02698-8

Bourgeois, F. T., Murthy, S., Pinto, C., Olson, K. L., Ioannidis, J. P. A., & Mandl, K. D. (2012). Pediatric versus adult drug trials for conditions with high pediatric disease burden. *Pediatrics, 130*(2), 285–292. https://doi.org/10.1542/peds.2012-0139

Caldwell, P. H. Y., Murphy, S. B., Butow, P. N., & Craig, J. C. (2004). Clinical trials in children. *The Lancet, 364*(9436), 803–811. https://doi.org/10.1016/S0140-6736(04)16942-0

Carpenter, D., Gonzalez, D., Retsch-Bogart, G., Sleath, B., & Wilfond, B. (2017). Methodological and ethical issues in pediatric medication safety research. *Pediatrics, 140*(3), e20170195. https://doi.org/10.1542/peds.2017-0195

Conroy, S., Choonara, I., Impicciatore, P., Mohn, A., Arnell, H., Rane, A., Knoeppel, C., Seyberth, H., Pandolfini, C., Raffaelli, M. P., Rocchi, F., Bonati, M., ’t Jong, G. W., de Hoog, M., & van den Anker, J. N. (2000). Survey of unlicensed and off label drug use in paediatric wards in European countries. *BMJ, 320*(7227), 79–82. https://doi.org/10.1136/bmj.320.7227.79

Contopoulos-Ioannidis, D. G., Baltogianni, M. S., & Ioannidis, J. P. A. (2010). Comparative effectiveness of medical interventions in adults versus children. *The Journal of Pediatrics, 157*(2), 322–330.e17. https://doi.org/10.1016/j.jpeds.2010.02.011

Costa, H. T. M. de L., Costa, T. X., Martins, R. R., & Oliveira, A. G. (2018). Use of off-label and unlicensed medicines in neonatal intensive care. *PLOS ONE, 13*(9), e0204427. https://doi.org/10.1371/journal.pone.0204427

Cuzzolin, L., Atzei, A., & Fanos, V. (2006). Off-label and unlicensed prescribing for newborns and children in different settings: A review of the literature and a consideration about drug safety. *Expert Opinion on Drug Safety, 5*(5), 703–718. https://doi.org/10.1517/14740338.5.5.703

Dean, P. H., Carr, R., Jensen, C., & Strahlendorf, C. (2021). Navigating the complex realm of drug access in paediatric care. *Paediatrics & Child Health, 26*(6), 334–336. https://doi.org/10.1093/pch/pxab001

Domingues, C., Jarak, I., Veiga, F., Dourado, M., & Figueiras, A. (2023). Pediatric drug development: Reviewing challenges and opportunities by tracking innovative therapies. *Pharmaceutics, 15*(10), Article 2431. https://doi.org/10.3390/pharmaceutics15102431

Dunne, J., Rodriguez, W. J., Murphy, M. D., Beasley, B. N., Burckart, G. J., Filie, J. D., Lewis, L. L., Sachs, H. C., Sheridan, P. H., Starke, P., & Yao, L. P. (2011). Extrapolation of adult data and other data in pediatric drug-development programs. *Pediatrics, 128*(5), e1242–e1249. https://doi.org/10.1542/peds.2010-3487

Ivanovska, V., Rademaker, C. M. A., van Dijk, L., & Mantel-Teeuwisse, A. K. (2014). Pediatric drug formulations: A review of challenges and progress. *Pediatrics, 134*(2), 361–372. https://doi.org/10.1542/peds.2013-3225

Kearns, G. L., Abdel-Rahman, S. M., Alander, S. W., Blowey, D. L., Leeder, J. S., & Kauffman, R. E. (2003). Developmental pharmacology—Drug disposition, action, and therapy in infants and children. *New England Journal of Medicine, 349*(12), 1157–1167. https://doi.org/10.1056/NEJMra035092

Kern, S. E. (2009). Challenges in conducting clinical trials in children: Approaches for improving performance. *Expert Review of Clinical Pharmacology, 2*(6), 609–617. https://doi.org/10.1586/ecp.09.40

Kriegel, C., Festag, M., Kishore, R. S. K., Roethlisberger, D., & Schmitt, G. (2020). Pediatric safety of polysorbates in drug formulations. *Children, 7*(1), Article 1. https://doi.org/10.3390/children7010001

Lagler, F. B., Hirschfeld, S., & Kindblom, J. M. (2021). Challenges in clinical trials for children and young people. *Archives of Disease in Childhood, 106*(4), 321–325. https://doi.org/10.1136/archdischild-2019-318676

Lathyris, D., Panagiotou, O. A., Baltogianni, M., Ioannidis, J. P. A., & Contopoulos-Ioannidis, D. G. (2014). Safety of medical interventions in children versus adults. *Pediatrics, 133*(3), e666–e673. https://doi.org/10.1542/peds.2013-3128

Litalien, C., Autmizguine, J., Carli, A., Giroux, D., Lebel, D., Leclerc, J.-M., Théorêt, Y., Gilpin, A., & Bérubé, S. (2020). Providing suitable pediatric formulations for Canadian children: A call for action. *The Canadian Journal of Hospital Pharmacy, 73*(4), 247–256. https://doi.org/10.4212/cjhp.v73i4.3023

Meng, M., Lv, M., Wang, L., Yang, B., Jiao, P., Lei, W., Lan, H., Shen, Q., Luo, X., Zhou, Q., Yu, X., Xun, Y., Lei, R., Hou, T., Chen, Y., & Li, Q. (2022). Off-label use of drugs in pediatrics: A scoping review. *European Journal of Pediatrics, 181*, 3259–3269. https://doi.org/10.1007/s00431-022-04515-7

Milne, C.-P., & Bruss, J. B. (2008). The economics of pediatric formulation development for off-patent drugs. *Clinical Therapeutics, 30*(11), 2133–2145. https://doi.org/10.1016/j.clinthera.2008.11.019

Naka, F., Strober, B., & Shahriari, M. (2017). Clinical trials: Kids are not just little people. *Clinics in Dermatology, 35*(6), 583–593. https://doi.org/10.1016/j.clindermatol.2017.08.008

Nebeska, K., Souckova, L., Stepanova, R., & Demlova, R. (2025). Exploring the publication gap in pediatric randomized clinical trials: Completed vs. uncompleted pediatric clinical trials. *Frontiers in Medicine, 12*, Article 1590125. https://doi.org/10.3389/fmed.2025.1590125

Nellis, G., Metsvaht, T., Varendi, H., Toompere, K., Lass, J., Mesek, I., Nunn, A. J., Turner, M. A., & Lutsar, I. (2015). Potentially harmful excipients in neonatal medicines: A pan-European observational study. *Archives of Disease in Childhood, 100*(7), 694–699. https://doi.org/10.1136/archdischild-2014-307793

O'Hara, K., Martin, J. H., & Schneider, J. J. (2020). Barriers and challenges in performing pharmacokinetic studies to inform dosing in the neonatal population. *Pharmacy, 8*(1), Article 16. https://doi.org/10.3390/pharmacy8010016

Petkova, V., Georgieva, D., Dimitrov, M., & Nikolova, I. (2023). Off-label prescribing in pediatric population—Literature review for 2012–2022. *Pharmaceutics, 15*(12), Article 2652. https://doi.org/10.3390/pharmaceutics15122652

Pica, N., & Bourgeois, F. T. (2016). Discontinuation and nonpublication of randomized clinical trials conducted in children. *Pediatrics, 138*(3), e20160223. https://doi.org/10.1542/peds.2016-0223

Raja, P., Duffett, M., Mazer-Amirshahi, M., Patel, A., Gilpin, A., Litalien, C., Chan, A. K., van den Anker, J., Lacaze-Masmonteil, T., & Samiee-Zafarghandy, S. (2020). Pediatric drug data in Canadian drug monographs: A descriptive analysis. *CMAJ Open, 8*(3), E522–E529. https://doi.org/10.9778/cmajo.20200010

Samuels, S., Park, K., Bhatt-Mehta, V., Sun, H., Mulugeta, Y., Yao, L., Green, D. J., & Burckart, G. J. (2023). Pediatric efficacy extrapolation in drug development submitted to the US Food and Drug Administration 2015–2020. *The Journal of Clinical Pharmacology, 63*(3), 307–313. https://doi.org/10.1002/jcph.2160

Singh, K., Franson, T., McCune, S., Jorgensen, D., Getz, K., Bearer, C., & Davis, J. M. (2025). Breaking the silence: Challenges and opportunities in pediatric drug development. *Pediatric Research, 98*, 807–812. https://doi.org/10.1038/s41390-025-03923-3

van der Zanden, T. M., Mooij, M. G., Vet, N. J., Neubert, A., Rascher, W., Lagler, F. B., Male, C., Grytli, H., Halvorsen, T., de Hoog, M., & de Wildt, S. N. (2021). Benefit-risk assessment of off-label drug use in children: The BRAvO framework. *Clinical Pharmacology & Therapeutics, 110*(4), 952–965. https://doi.org/10.1002/cpt.2336

van der Zanden, T. M., Smeets, N. J. L., de Hoop-Sommen, M., Schwerzel, M. F. T., Huang, H. J., Barten, L. J. C., van der Heijden, J. E. M., Freriksen, J. J. M., Horstink, A. A. L., Holsappel, I. H. G., Mooij, M. G., de Hoog, M., & de Wildt, S. N. (2022). Off-label, but on-evidence? A review of the level of evidence for pediatric pharmacotherapy. *Clinical Pharmacology & Therapeutics, 112*(6), 1243–1253. https://doi.org/10.1002/cpt.2736 van Riet-Nales, D. A., de Jager, K. E., Schobben, A. F. A. M., Egberts, T. C. G., & Rademaker, C. M. A. (2011). The availability and age-appropriateness of medicines authorized for children in the Netherlands. *British Journal of Clinical Pharmacology, 72*(3), 465–473. https://doi.org/10.1111/j.1365-2125.2011.03982.x

van Riet-Nales, D. A., de Neef, B. J., Schobben, A. F. A. M., Ferreira, J. A., Egberts, T. C. G., & Rademaker, C. M. A. (2013). Acceptability of different oral formulations in infants and preschool children. *Archives of Disease in Childhood, 98*(9), 725–731. https://doi.org/10.1136/archdischild-2012-303303

Watanabe, H., Nagano, N., Tsuji, Y., Noto, N., Ayusawa, M., & Morioka, I. (2024). Challenges of pediatric pharmacotherapy: A narrative review of pharmacokinetics, pharmacodynamics, and pharmacogenetics. *European Journal of Clinical Pharmacology, 80*(2), 203–221. https://doi.org/10.1007/s00228-023-03598-x

Weiler, H. R., Le, L. M., & Fadda, H. M. (2026). Age-appropriateness of oral formulations authorized for pediatrics. *Pharmaceutical Research, 43*, 1785–1797. https://doi.org/10.1007/s11095-026-04055-x